

PHASE 5 • DEPLOYMENT

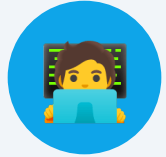
Explainable AI-Powered Type 2 Diabetes Risk Prediction Platform

CDC BRFSS 2023 • CatBoost • SHAP & LIME • Full-Stack Deployment

Data Science Capstone • July 2026

CONTRIBUTORS

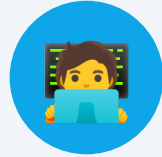
The Team



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Modeling

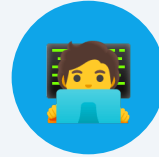
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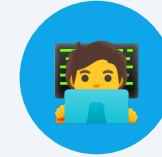
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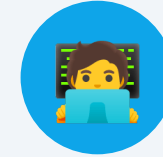
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THE PROBLEM

A Global Health Crisis

537M

People affected globally

50%

Remain undiagnosed

70%

Preventable with early action

Source: International Diabetes Federation, IDF Diabetes Atlas (10th ed., 2021)

Why Screening Falls Short



Screening requires hospital visits and laboratory tests



Online calculators return a number with no explanation



Users never understand why they received a given risk score



Late diagnosis leads to preventable, irreversible complications

An Explainable AI Platform



Predicts

Type 2 diabetes risk from 14
self-reportable health
questions



Explains

Every prediction with SHAP
and LIME — no black box



Prevents

Promotes awareness and
lifestyle intervention early



Supports

Informed, data-backed clinical
discussions

Six Objectives Achieved

- ✓ Clean and prepare CDC BRFSS 2023 survey data (429,086 respondents) for ML
- ✓ Explore demographic, behavioural, and clinical risk factors
- ✓ Train and compare Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost
- ✓ Select CatBoost and tune its decision threshold to prioritise recall
- ✓ Implement SHAP and LIME for global and per-patient explainability
- ✓ Ship a deployed, full-stack dashboard — not just a notebook

Dataset at a Glance

429,086

Survey respondents

14

Selected features

2023

BRFSS survey year

0

Lab tests required

Feature categories (fourteen self-reportable inputs — no clinic visit required):

Anthropometric

BMI

Demographic

Age, sex, race

Health status

General health, physical-health days

Behavioural

Smoking, physical activity

Socio-economic

Education, income

Clinical history

Hypertension, cholesterol, kidney disease, heart disease

HOW IT'S BUILT

System Architecture

Presentation	Next.js 14 dashboard + Dida assistant
Application / API	FastAPI — predict, batch, export, chat
ML & Explainability	CatBoost + SHAP TreeExplainer + LIME
Data & Identity	PostgreSQL · Google OAuth 2.0 · JWT
Reporting	PDF (ReportLab) · CSV · SMTP email

All three interaction surfaces — the structured form, batch upload, and Dida — call the same /predict endpoint: a single source of truth for risk computation.

Technology Stack



Next.js 14

Frontend — App Router



FastAPI

Async Python backend



PostgreSQL

Async SQLAlchemy ORM



CatBoost

Gradient boosting model



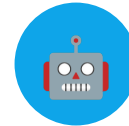
SHAP + LIME

Dual explainability layer



Google OAuth

NextAuth + JWT bearer



Llama-3.1-8B

Dida's conversation layer



ReportLab

PDF report generation

Why a Full Web App, Not a Streamlit Demo

This Platform		Streamlit Demo
Secure authentication	✓	Limited
User accounts & history	✓	✗
Explainability dashboards	✓	Basic
Conversational AI assistant	✓	Limited
Mobile responsive	✓	Poor
Production scalability	✓	Limited

Deployment Topology

Frontend	Next.js 14	Node.js SSR hosting
Backend API	FastAPI (async)	ASGI via uvicorn, containerised
Database	PostgreSQL	Managed instance
Model artifacts	CatBoost + preprocessor	Loaded from disk at API startup
Conversational AI	Llama-3.1-8B-Instruct	External inference API (NVIDIA-hosted)
Email delivery	SMTP	Outbound app-password auth

Secrets (SMTP credentials, OAuth client secret, LLM API key) are supplied via environment variables — never committed to source control.

Designed for Real Users

Primary Stakeholders

- Clinicians & healthcare providers
- Patients & individuals at risk
- Public health agencies
- Health promotion programmes

Secondary Stakeholders

- Researchers
- Academic institutions
- Health insurance organisations
- Policy makers & NGOs

Why it matters: anyone with a browser can access the platform — no installation, no programming knowledge required.

FEATURES

Complete AI Health Platform



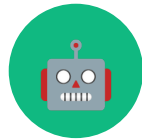
User

Google Authentication

Personalised Dashboard

Risk Prediction

Prediction History



AI

Diabetes Prediction

SHAP Explainability

LIME Explanations

Dida Health Assistant



Education

Learning Centre

Prevention Guides

Symptoms & Risk Factors

Disease Progression



Research

Literature Review

Dataset Explorer

Model Card

XAI Documentation

From Login to Insight



RESULTS

Model Performance

75%

Recall (sensitivity)

Prioritised — fewer missed at-risk respondents

82%

ROC-AUC

Strong overall discriminative power

Held-out test set • CDC BRFSS 2023 • CatBoost with recall-optimised decision threshold (no synthetic oversampling)

Global SHAP analysis consistently ranks general health status, BMI, age group, and hypertension as the strongest predictors — matching established clinical risk factors.

Every Prediction is Explained

SHAP

Shows which features increased or reduced predicted diabetes risk across the entire model.

Global explanations

Feature ranking

Direction of impact

LIME

Explains why this individual prediction was made by locally perturbing features.

Local explanations

Per-prediction

Human-readable

Meet Dida



Hybrid architecture: an LLM for open conversation, a deterministic state machine for the actual assessment.

My BMI is 32 and I have high blood pressure. What's my risk?

Let's find out precisely — I'll walk you through 14 quick questions and give you a full explanation, not just a number.

What does my SHAP output mean?

Your BMI contributed most to the positive prediction (+0.18). Physical activity reduced your risk (-0.06) — the two factors you can most influence.

Benefits of the Platform

For Patients

Easy risk screening

Better understanding

Early awareness

Personalised education

For Clinicians

Explainable predictions

Faster screening

Better communication

For Public Health

Preventive healthcare

Improved accessibility

Lifestyle interventions

Future Enhancements

- Social media bots (WhatsApp, Telegram)

- Wearable device integration

- Multi-language support

- Doctor referral system

- EHR integration

- Continuous glucose monitoring

- Native mobile app

- Population analytics dashboard

CONCLUSION

Key Takeaways

- 01** Explainable AI makes predictions transparent and trustworthy
- 02** The platform combines prediction, education, and prevention in one place
- 03** A deployed web app delivers far greater accessibility than a notebook or demo
- 04** Dida's hybrid design proves LLMs and deterministic pipelines work best together

Thank You · Questions?